



Research article

Variable-coupled distributed resource allocations for multi-agent system with limited admissible interaction

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ABSTRACT

Variable-coupled distributed resource allocations (VCDRA) involve optimizing resource distribution among interconnected entities, reflecting the complex correlation in practical systems. Based on the consensus technology, this paper proposes a novel distributed optimization algorithm to address the VCDRA in scenarios where agents are confronted with admissible interaction ranges and external disturbances. Admissible interaction refers to agents only transmitting information within a specified range due to limited communication capabilities or resource availability. The proposed distributed optimization algorithm improves existing interaction mechanisms, reducing computational demands and unifying the communication schemes of the multi-agent system, thereby avoiding additional information exchange. Theoretically, it is proven that the algorithm can get the optimal solution of VCDRA with an exponential rate. Compared to existing algorithms for suppressing external disturbances, the robustness of the proposed distributed optimization algorithm no longer relies on the upper bound of external disturbances, allowing it to remain effective even in the presence of unbounded disturbances. Finally, smart grids and wireless communications applications demonstrate the convergence and robustness of the developed distributed optimization algorithm, further proving its superiority in practical applications.

1. Introduction

Distributed resource allocations (DRAs), involving the optimization of allocation cost across multiple entities, have garnered considerable attention in fields such as economic dispatch [1], wireless sensor networks [2], and energy management [3]. These problems are critical in contexts requiring efficient resource use to meet operational needs and achieve optimal system performance.

Recently, the authors of [4] integrated solar photovoltaic systems with cogeneration power plants to enhance efficiency and reduce operational costs. Performance assessments using multiple-criteria decision-making models have shown the advantages of different sustainable energy systems [5]. Bioinspired techniques such as the jellyfish optimization algorithm have proven effective for tracking maximum power points in solar photovoltaic systems under varying environmental conditions [6]. While these approaches have demonstrated significant benefits, they often fall short in scenarios requiring real-time adaptability and privacy preservation [7]. Therefore, distributed optimization algorithms with real-time feedback and privacy protection have been found to solve the DRA well by utilizing the cooperative operation mechanism of multi-agent systems [8,9]. Inspired by the pioneering work of DRA [10], various distributed optimization algorithms have

been developed. The authors of [11] applied a distributed optimization algorithm based on the weighted gradient method to solve DRAs. A primal-dual distributed optimization algorithm was proposed by utilizing dynamic average consensus to estimate coupled constraints [12]. Addressing equipment limitations and generating capacity, DRAs with local feasible constraints were solved by using a projected distributed optimization algorithm in [13] and a projection-gradient distributed optimization algorithm in [14]. Although the authors of [15,16] have successfully developed effective distributed optimization algorithms to address variable-coupled inequality constraints and analyzed their convergence, practical applications often encounter scenarios where the local cost function of an agent is influenced by others. For example, in the power dispatching of wind farms [17], the operation of each wind turbine can affect the performance of neighboring turbines due to wind flow interactions. Additionally, scenarios such as multi-regional economic coordinated dispatch [18] and distributed model predictive control in intelligent energy systems [19] often exhibit strong coupling relationships between devices. Hence, there is significant value in designing algorithms capable of solving DRAs with variable-coupled private cost functions.

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Such a kind of DRA is the variable-coupled DRA (VCDRA). So far, Firouzbahrami and Nobakhti have introduced finite-time distributed optimization algorithms for solving VCDRA in [20,21]. However, the effectiveness of these algorithms is only guaranteed under static communication networks. This limitation restricts their applicability in scenarios where communication networks vary over time. Although the authors of [22,23] respectively discussed a distributed online algorithm for open multi-agent systems to solve the distributed resource allocation in a dynamic network setting, in many practical applications, the admissible interaction range plays a crucial role. For example, the operation of a wind farm where generators interact with each other within a limited range [17], or a satellite system where satellites communicate with neighboring satellites on adjacent orbital planes [24]. This setting inevitably causes the communication link to exhibit time-varying characteristics based on the real-time state of the agent. It is important to emphasize that the distributed optimization algorithms in [20,21] are not suitable for multi-agent systems with variable interaction ranges. Consequently, there remains a significant research gap in addressing VCDRA under multi-agent systems that are equipped with admissible interaction ranges.

In scenarios where multi-agent systems are affected by external factors such as communication noise, environmental interference, and equipment vibrations, these disturbances can significantly degrade system performance or lead to failures. Therefore, effectively overcoming disturbance effects is essential to enhance system reliability and improve performance. Recent years have seen the development of distributed optimization algorithms capable of handling external disturbances, including internal model-based distributed optimization algorithms in [25,26] for solving DRAs under known disturbance, a reduced-order extended distributed optimization algorithm based on the active disturbance approach in [27] under unknown cases, and a high-order distributed optimization algorithm based on disturbance observer in [28]. However, these algorithms typically rely on upper bounds of disturbances for their application. Recognizing this limitation, it is crucial to introduce anti-disturbance distributed optimization algorithms tailored to VCDRA without using the disturbance information for better robustness and adaptability.

The previous discussion and existing research gap drive this paper to develop an anti-disturbance distributed optimization algorithm for the VCDRA under an admissible interaction setting. In comparison with existing literature, the primary contributions of this paper are summarized as follows.

- Compared to [14,16,29], this paper considers variable-coupled private cost functions to better capture coupling among agents. Moreover, the considered VCDRA significantly broadens the applicability of the previous work [30,31], allowing for scenarios where strong convexity may not universally hold across all private cost functions.
- In view of practical limitations in sensing and signal coverage, which are overlooked in multi-agent systems discussed in [20,21], this paper integrates the admissible interaction range into distributed optimization algorithms for VCDRA. Compared to [32], the proposed interaction mechanism unifies the communication topology within the controller, thereby reducing computational burden and eliminating additional private information exchanges.
- The distributed optimization algorithm proposed in this paper achieves exponential convergence of the optimal solution of VCDRA without relying on explicit upper bounds of external disturbances, which makes it have superior robustness than the anti-disturbance algorithms in [21,25–28].

Here is a brief overview of this paper. Section 2 provides relevant symbols and background information. In Section 3, the considered multi-agent system and VCDRA are introduced, followed by the development of a distributed optimization algorithm. The theoretical

analysis of the proposed algorithm is discussed in Section 4. Section 5 presents numerical simulations based on real-world applications, providing data support for the theoretical results of this paper. Finally, the last section summarizes the paper.

2. Preliminaries

This section introduces essential notations and lemmas relevant to the content of this paper. The purpose here is to establish a foundational understanding necessary for subsequent discussions.

For the real space $\mathbb{R}$, $\mathbb{R}_+^n$ denotes the n -dimensional positive real vector set. $\mathbb{R}^{m \times n}$ is the collection of $m \times n$ -dimensional real matrix. For $m \geq 1$, $J_m = \{1, 2, \dots, m\}$ is a finite integer set. $(\cdot)^\top$ means the transpose, and $a_{(n)} = (a, a, \dots, a)^\top \in \mathbb{R}^n$ for any $a \in \mathbb{R}$. $\otimes$ is the Kronecker product, and $\text{col}(\cdot)$ is the vector stacking. The indicator function of a set $S \subseteq \mathbb{R}^n$ is denoted by

$$I_S(x) = \begin{cases} 1, & x \in S \\ 0, & x \notin S, \end{cases}$$

where $x \in \mathbb{R}^n$. Denote $D^-f(x) = \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$ as the left Dini derivative of f at x .

The multi-agent system discussed in this paper has a limited allowable interaction range, so the communication topology of the multi-agent system changes with time. Therefore, for time-varying communication topology, it is generally denoted as $\mathcal{G}_{A(t)} = (\mathcal{J}_n, \mathcal{E}_{A(t)}, A(t))$, where $\mathcal{J}_n$ is node set, $\mathcal{E}_{A(t)} = \{(i, j) \in \mathcal{J}_n \times \mathcal{J}_n : a_{ij}(t) > 0\}$ is edge set, and $A(t) = [a_{ij}(t)]_{n \times n}$ is adjacency matrix. For all $t \geq 0$, we assume $a_{ii}(t) = 0$ in this paper ($\forall i \in \mathcal{J}_n$). Let $\mathcal{N}_i(t) = \{j \in \mathcal{J}_n : a_{ij}(t) > 0\}$ be the neighbor set of node i at time t , then $\mathcal{G}_{A(t)}$ is undirected if and only if $j \in \mathcal{N}_i(t)$ is equivalent to $i \in \mathcal{N}_j(t)$. The undirected graph $\mathcal{G}_{A(t)}$ is connected if there is a connected line between any two nodes in the graph. The Laplacian matrix of the graph $\mathcal{G}_{A(t)}$ is defined as $\mathcal{L}_{A(t)} = E(t) - A(t)$ with $E(t) = \text{diag}\{\sum_{j=1}^n a_{ij}(t)\}$.

Lemma 2.1 ([33]). For an undirected and connected graph $\mathcal{G}_A = (\mathcal{J}_n, \mathcal{E}_A, A)$ with $A = [a_{ij}]_{n \times n}$, its Laplacian matrix $\mathcal{L}_A$ is positive semi-definite, and $s^\top \mathcal{L}_A s = \frac{1}{2} \sum_{i,j=1}^n a_{ij} (s_i - s_j)^2$ for all $s = \text{col}(s_i) \in \mathbb{R}^n$. Furthermore, all eigenvalues of $\mathcal{L}_A$ are $0 = \lambda_1(\mathcal{L}_A) < \lambda_2(\mathcal{L}_A) \leq \dots \leq \lambda_n(\mathcal{L}_A)$, and $\lambda_2(\mathcal{L}_A) s^\top \mathcal{L}_A s \leq s^\top \mathcal{L}_A \mathcal{L}_A s$ holds for any $s \in \mathbb{R}^n$.

Definition 2.1 ([34]). If there is $\mu > 0$ such that $f(s) - f(p) \geq \langle \nabla f(p), s - p \rangle + \frac{\mu}{2} \|s - p\|^2$ holds for any $p, s \in \mathbb{R}^n$, the function f is μ -strongly convex.

Lemma 2.2 ([35]). Consider the system $\dot{s} = P(s)$ with $s(0) = s_0$. Assume that the origin is an equilibrium point, if there exists a continuous function $V(s) \geq 0$ satisfying $\lim_{\|s\| \rightarrow \infty} V(s) = \infty$, $V(s) = 0$ if and only if $s = 0$, and $\dot{V}(s) \leq -\alpha V(s)$ with $\alpha > 0$, then the origin is globally exponentially stable.

3. Problem formulation and algorithm design

In this section, a multi-agent system equipped with an admissible interaction range and the variable-coupled distributed resource allocation (VCDRA) are introduced. The objective of this section is to propose an interaction mechanism and a distributed optimization algorithm for addressing VCDRA in the presence of external disturbances.

Specifically, this paper considers a group of n agents, where the dynamic of agent $i \in \mathcal{J}_n$ is described by the following system model

$$\dot{y}_i = u_i + g(t), \quad x_i = o_i(y), \quad (1)$$

where $y_i \in \mathbb{R}$ is the i th agent's state, $u_i \in \mathbb{R}$ is the control input, $g(t) \in \mathbb{R}$ denotes the time-varying external disturbance, and $x_i \in \mathbb{R}$ is the output that depends on the global state $y = \text{col}(y_i)_{i=1}^n$, and o_i denotes a nonlinear vector function.

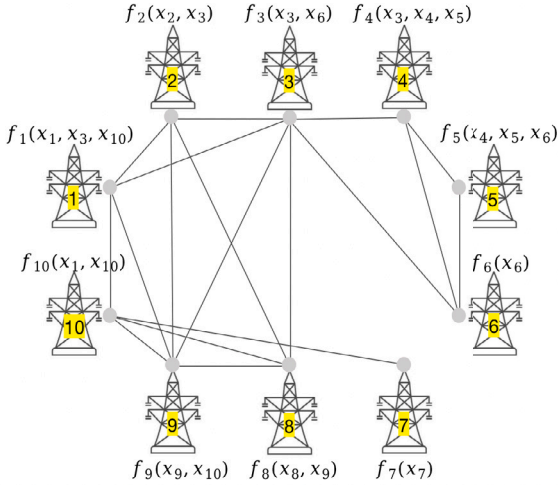


Fig. 1. Connected network of agents in problem (2).

For each agent in the network, a private cost function $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and a private resource demand $d_i \in \mathbb{R}$ are assigned. The aim of this paper is to propose a distributed optimization algorithm to ensure agents collaboratively solve the following VCDRA with the resource-balance constraint

$$\begin{aligned} \min \quad & F(x) = \sum_{i=1}^n f_i(x_i, x_{\mathcal{N}_i^I}) \\ \text{s.t.} \quad & \sum_{i=1}^n x_i = \sum_{i=1}^n d_i, \end{aligned} \quad (2)$$

where $x = \text{col}(x_i)_{i=1}^n \in \mathbb{R}^n$ is the decision variable, the objective function $F(x)$ represents the total cost of resource allocation in the multi-agent system, $\mathcal{N}_i^I \subseteq \mathcal{J}_n$ is the index set of the independent variables excluding x_i of f_i , and $x_{\mathcal{N}_i^I} = \{x_j \in \mathbb{R} : j \in \mathcal{N}_i^I\}$.

Remark 3.1. In VCDRA (2), the private cost functions $f_i(x)$ are variable-coupled, representing a broader case compared to DRAs discussed in [36–38]. Such problems arise in various fields including wireless transmission [39], multi-area economic dispatch [18], and multi-microgrid systems [40].

For instance, in wireless transmission, the data transmission rate between different nodes may be affected by competition for shared resources. Consequently, the private cost function for each device depends not only on its own state but also on the state of neighboring devices. Fig. 1 illustrates a simple network to depict this scenario. Take the device labeled as 1 as example, it sends and receive information with devices 2, 3, 9, and 10, but shares the same type of resource with devices 3 and 10. In this case, besides its own influence, the generation cost f_1 is also affected by devices 3 and 10. Therefore, it has $\mathcal{N}_1^I = \{3, 10\}$ and the private cost function of the device 1 is denoted as $f_1(x_1, x_3, x_{10})$ instead of $f_1(x_1)$.

Assumption 1. The team cost function $F(x)$ is strongly convex.

According to Assumption 1, only the strong convexity of the team cost function is necessary, relaxing the strong convexity requirement for all private cost functions as discussed in [41, 42].

Lemma 3.1 ([43]). Under Assumption 1, $x^* \in \mathbb{R}^n$ is an optimal solution to VCDRA (2) if and only if there is $c \in \mathbb{R}$ such that

$$\begin{cases} \sum_{i=1}^n x_i^* = \sum_{i=1}^n d_i, & (a) \\ \nabla F(x^*) = c_{(n)} & (b). \end{cases} \quad (3)$$

Due to the inherent limitation of the interaction abilities of agents in the multi-agent system (1), in reality, each agent establishes communication links for information exchange only when its state $y_i(t)$ falls within a specific range relative to other agents. By employing a central unit connected to all agents to assess their interactive capabilities, we can determine that the neighbor set of agent i at time t is defined as

$$\mathcal{N}_i(t) = \{j \in \mathcal{J}_n : |y_i(t) - y_j(t)| < R\},$$

where $R > 0$ represents the maximum interaction range of the system (1).

Assumption 2. For the multi-agent system (1), the maximal interaction range $R > 0$ satisfies $\mathcal{N}_i^I \subseteq \mathcal{N}_i(0)$ for all $i \in \mathcal{J}_n$.

It should be noted that agents have the potential to affect each other's private cost functions only if they are connected to each other, so Assumption 2 is natural from a practical scenario perspective.

To address VCDRA (2) under the above admission interaction range, based on Lemma 3.1 and the gradient consensus technology, the following distributed optimization algorithm over the system (1) is designed.

$$\begin{cases} \dot{x}_i(t) = \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) + d_i \\ \dot{y}_i(t) = - \sum_{j=1}^n w_{ij}(t)(\eta_i(t) - \eta_j(t)) + g(t) \\ \eta_i(t) = \frac{\partial}{\partial x_i} f_i(x_i(t), x_{\mathcal{N}_i^I}(t)) + \sum_{j \in \mathcal{N}_i^I(0)} \frac{\partial}{\partial x_i} f_j(x_j(t), x_{\mathcal{N}_j^I}(t)) \end{cases} \quad (4)$$

where $g(t)$ is the external disturbance, and the interaction mechanism is designed as

$$w_{ij}(t) = w(|y_i(t) - y_j(t)|) = \begin{cases} 0, & \text{if } j \notin \mathcal{N}_i(0) \text{ and } j \notin \mathcal{N}_i(t) \\ 1, & \text{if } j \notin \mathcal{N}_i(0) \text{ and } j \in \mathcal{N}_i(t) \\ v(|y_i(t) - y_j(t)|), & \text{if } j \in \mathcal{N}_i(0) \end{cases} \quad (5)$$

with $v(s) = I_{[0, \tau R)}(s) + \frac{(1-\tau^2)R^2}{R^2-s^2} I_{[\tau R, R)}(s)$ for $R > 0$ and $\tau \in (0, 1)$. Generally, τ can be seen as the lossless transmission capability of the sensor or the level of vigilance set by the agent when preventing a collision.

Remark 3.2. The proposed distributed optimization algorithm (4) is based on the gradient consensus technology with remarkable efficiency, scalability, and anti-disturbance ability in multi-agent systems. First, the gradient consensus technology ensures that the state solution of the algorithm (4) converges to the exact optimal solution, which reflects the efficiency of the algorithm in solving VCDRA. Secondly, the distributed operation mechanism allows the multi-agent system to maintain stable performance when adding or removing agents, which reflects the scalability of the algorithm (4). Finally, due to the relativity of the gradient consensus technique, the proposed algorithm shows strong robustness to unknown external disturbances. This enables the algorithm (4) to effectively allocate resources even when the disturbance bound is unknown. Specifically, the operating mechanism of the algorithm (4) is shown in Fig. 2.

Lemma 3.2 ([32]). Denote $\mathcal{G}_{A(t)} = (\mathcal{J}_n, \mathcal{E}_{A(t)}, A(t))$ ($t \geq 0$) as a time-varying undirected graph, and $\mathcal{L}_{A(t)}$ as its Laplacian matrix. If $\mathcal{G}_{A(0)}$ is connected and $\mathcal{E}_{A(0)} \subseteq \mathcal{E}_{A(t)}$, then it has $\lambda_2(\mathcal{L}_{A(t)}) \geq \lambda_2(\mathcal{L}_{A(0)})$ for all $t \geq 0$.

Remark 3.3. It is important to emphasize that the distributed optimization algorithm (4) represents more than just an extension of algorithms designed for existing fixed undirected communication topologies. The definition of the interaction mechanism (5) implies that

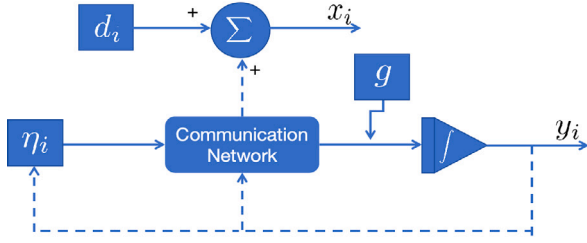


Fig. 2. The operating framework of the distributed optimization algorithm (4).

the communication topology $\mathcal{G}_{W(t)}$ is both time-varying and state-dependent. This characteristic significantly increases the complexity of analyzing the convergence properties of the distributed optimization algorithm (4), compared to the fixed communication networks studied in [20,21], and so on.

Remark 3.4. It can be seen from the proposed algorithm (4) that the terms $\frac{\partial}{\partial x_i} f_j(x_j(t), x_{N_j^i}(t))$ ($j \in \mathcal{N}_i(0)$) are utilized in the local operation of the i th agent. These terms reflect the impact of agent i on the private cost function of agent j , which is typically accessible to agent i in practical scenarios. Furthermore, to evaluate the interaction weights $w_{ij}(t)$ of the agents, it is natural to introduce a central processor to collect the global state y of the system (1).

It is important to note that the algorithm (4) still operates in a distributed manner under a given $w_{ij}(t)$. Therefore, it functions as a fully distributed optimization algorithm that does not rely on any global information, such as the connectivity degree of the communication network or artificially selected control parameters. The autonomy of system operation is further underscored by its flexibility in the selection of initial points. This flexibility represents a significant enhancement over the distributed optimization algorithms discussed in [20,29], where specific initial conditions, such as $\sum_{i=1}^n x_i(0) = \sum_{i=1}^n d_i$ or $x_i(0)$ being the optimal solution to the private cost function f_i , are required.

Remark 3.5. In the distributed optimization algorithm (4), the external disturbances $g(t)$ affecting the agents are synchronized. This scenario arises when the entire network is influenced by the same noise or attack. Importantly, even if the external disturbance $g(t)$ becomes unbounded, the gradient consensus technique used in the algorithm ensures the control input and output states of each agent remain bounded, thus preventing excessive energy consumption.

Subsequent analysis will demonstrate that the effectiveness of the algorithm (4) does not depend on the boundedness of $g(t)$. This characteristic renders it more general than existing disturbance rejection distributed optimization algorithms in [25–27].

4. Theoretical analysis

This section focuses on the theoretical analysis of the proposed distributed optimization algorithm (4) in terms of preserving connectivity and exponential convergence.

4.1. Connectivity-preserving analysis

Before proving the convergence of distributed optimization algorithm (4), we prove connectivity-preserving performance of the given interaction mechanism (5).

Define the following functions

$$K_1(t) = \sum_{i,j=1}^n \int_0^{|y_i(t)-y_j(t)|} w(s)ds, \quad K_2(t) = y(t)^T \mathcal{L}_{W(t)} y(t), \quad (6)$$

and

$$K(t) = K_1(t) - K_2(t), \quad (7)$$

where $y(t) = \text{col}(y_i(t))$. Then it deduces the following connectivity-preserving performance of the distributed optimization algorithm (4).

Theorem 4.1. Under Assumptions 1–2, if $\mathcal{G}_{W(0)}$ is connected and $K(0)$ is bounded, then $\mathcal{E}_{W(0)} \subseteq \mathcal{E}_{W(t)}$ for all $t \geq 0$.

Proof. Assume that the conclusion does not hold, then there are $(i, j) \in \mathcal{E}_{W(0)}$ and $t_1 > 0$ such that the link between agents i and j breaks at t_1 for the first time, that is, $|y_i(t) - y_j(t)| < R$ for $t \in [0, t_1)$ and

$$\lim_{t \rightarrow t_1^-} |y_i(t) - y_j(t)| = R. \quad (8)$$

Based on the distributed optimization algorithm (4) and the definition of $K_1(t)$ in (6), it has

$$\begin{aligned} \dot{K}_1(t) &= \sum_{i,j=1}^n w_{ij}(t)(y_i(t) - y_j(t))(\dot{y}_i(t) - \dot{y}_j(t)) \\ &= - \sum_{i,j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) - \sum_{r=1}^n w_{jr}(t)(\eta_j(t) - \eta_r(t)) \right). \end{aligned} \quad (9)$$

Since $\mathcal{G}_{W(t)}$ is undirected for any $t \geq 0$, one gets

$$\begin{aligned} & - \sum_{i,j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) - \sum_{r=1}^n w_{jr}(t)(\eta_j(t) - \eta_r(t)) \right) \\ &= - \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) \right) \\ & \quad - \sum_{j=1}^n \left(\sum_{i=1}^n w_{ij}(t)(y_j(t) - y_i(t)) \right) \left(\sum_{r=1}^n w_{jr}(t)(\eta_j(t) - \eta_r(t)) \right) \\ &= -2 \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) \right). \end{aligned} \quad (10)$$

According to the definition of $v(s)$ in the interaction mechanism (5), it is easy to get that $v(s)$ is continuous and monotonously increased on $[0, R)$ and

$$\lim_{r \rightarrow R^-} \int_0^r v(s)ds = +\infty. \quad (11)$$

Hence, combining with (8), there is $\delta > 0$ such that $D^-w_{ij}(t) \geq 0$, $i, j \in \mathcal{J}_n$ for all $t \in (t_1 - \delta, t_1)$, which follows that $D^- \mathcal{L}_{W(t)}$ is a positive semi-definite matrix for all $t \in (t_1 - \delta, t_1)$. Therefore, for all $t \in (t_1 - \delta, t_1)$, the time derivative of $K_2(t)$ along the distributed optimization algorithm (4) yields

$$\begin{aligned} D^-K_2(t) &= 2y(t)^T \mathcal{L}_{W(t)} \dot{y}(t) + y(t)^T D^- \mathcal{L}_{W(t)} y(t) \\ &\geq 2 \sum_{i=1}^n \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \dot{y}_i(t) \\ &= 2 \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(- \sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) + g(t) \right). \end{aligned} \quad (12)$$

Noting that $w_{ij}(t) = w_{ji}(t)$ holds for all $t \geq 0$, it has $\sum_{i=1}^n \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) = 0$, and thus

$$\begin{aligned} D^-K_2(t) &\geq -2 \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) \right) \\ &\quad + 2g(t) \sum_{i=1}^n \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \\ &= -2 \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) \right) \\ &= \dot{K}_1(t). \end{aligned}$$

Therefore, $D^-K(t) = \dot{K}_1(t) - D^-K_2(t) \leq 0$ holds for all $t \in (t_1 - \delta, t_1)$, which implies that $K(t) \leq K(0) < +\infty$ for all $t \in (t_1 - \delta, t_1)$. Since

$y_i(t)$ is the state solution of the distributed optimization algorithm (4), we have $y_i(t)$ is absolutely continuous, thus $y_i(t)$ is bounded on $[0, t_1]$. Combining with the fact that $\mathcal{L}_{W(t)}$ is positive semi-definite, one has $0 \leq y(t)^\top \mathcal{L}_{W(t)} y(t) = K_2(t) \leq \lambda_n(\mathcal{L}_{W(t)}) \|y(t)\|^2 < +\infty$ for $t \in [0, t_1]$, that is $K_2(t)$ is bounded on $[0, t_1]$. Since $K_1(t) = K(t) + K_2(t) \leq K(0) + K_2(t)$, one gets $K_1(t)$ is bounded on $(t_1 - \delta, t_1)$. Besides, it follows from (11) that $\lim_{t \rightarrow t_1^-} K_1(t) = +\infty$, which leads to a contradiction. As a result, for all $t \geq 0$ and $(i, j) \in \mathcal{E}_{W(0)}$, $|y_i(t) - y_j(t)| < R$ holds, that is, $\mathcal{E}_{W(0)} \subseteq \mathcal{E}_{W(t)}$ for all $t \geq 0$. ■

Remark 4.1. The interaction mechanism (5) in this paper can be seen as an extension of the connectivity-preserving mechanism in [32], which shows obvious advantages in two aspects. On the one hand, the calculation time of the interactive mechanism in each operation process is changed from two to one, which greatly reduces the computational burden of the algorithm. On the other hand, the interaction mechanism (5) unifies the communication topology within the controller. This improvement avoids the additional exchange of the agent's private information and is particularly beneficial for privacy-sensitive applications.

4.2. Convergence analysis

Next, the convergence analysis of distributed optimization algorithm (4) is given.

Theorem 4.2. Under Assumptions 1–2, if $\mathcal{G}_{W(0)}$ is connected and $K(0)$ is bounded, then $x(t) = \text{col}(x_i(t))$ generated by the distributed optimization algorithm (4) exponentially converges to the optimal solution $x^* = \text{col}(x_i^*)$ to VCDRA (2).

Proof. Based on Theorem 4.1, it holds that $\mathcal{G}_{W(t)}$ is undirected and connected for any $t \geq 0$. Since $x_i(t) = \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) + d_i$, then we have $\sum_{i=1}^n x_i(t) = \sum_{i=1}^n \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) + \sum_{i=1}^n d_i = \sum_{i=1}^n d_i$, which means that $x(t) = \text{col}(x_i(t))$ satisfies (3a) for all $t \geq 0$. Thus, VCDRA (2) can be rewritten as

$$\min H(y(t)) \triangleq \sum_{i=1}^n f_i(\mathcal{L}_{W(t)} y(t) + d), \quad (13)$$

where $y(t) = \text{col}(y_i(t))$ and $d = \text{col}(d_i)$.

From the distributed optimization algorithm (4), there is

$$\frac{\partial x_i}{\partial y_j} = \begin{cases} \sum_{j=1}^n \varpi_{ij}(t) & j = i \\ -\varpi_{ij}(t) & j \neq i \end{cases}$$

where $\varpi_{ij}(t) = w(|y_i(t) - y_j(t)|)|y_i(t) - y_j(t)| + w_{ij}(t)$. Thus, by the Chain Rule [34] and $\nabla F(x) = \eta \triangleq \text{col}(\eta_i)$, it has

$$\nabla H(y) = \mathcal{L}_{A(t)} \nabla F(x), \quad (14)$$

with $\Lambda(t) = [\varpi_{ij}(t)]_{n \times n}$, and the Hessian matrix of $F(y)$ is

$$\nabla^2 H(y) = \mathcal{L}_{A(t)} \nabla^2 F(x) \mathcal{L}_{A(t)}. \quad (15)$$

Since $\mathcal{L}_{A(t)}$ is positive semi-definite and $\nabla^2 F(x)$ is positive definite, then $\nabla^2 H(y)$ is positive semi-definite, which implies that $H(y)$ is convex function.

Let $y^* = \text{col}(y_i^*)$ be the optimal solution to (13), then $x^* = \text{col}(x_i^*)$ with $x_i^* = \sum_{j=1}^n w_{ij}^*(y_i^* - y_j^*) + d_i$ and $w_{ij}^* = w(|y_i^* - y_j^*|)$ satisfies $\mathcal{L}_{A(t)} \nabla F(x^*) = \nabla H(y^*) = 0$, which implies that $\nabla F(x^*) = c \mathbf{1}_{(n)}$ for some $c \in \mathbb{R}$, that is, x^* is the optimal solution to VCDRA (2) from Lemma 3.1. For any $z_1, z_2 \in \mathbb{R}^n$, it follows from the Mean Value Theorem [43] that

$$H(z_1) = H(z_2) + \nabla H(z_2)^\top (z_1 - z_2) + \frac{1}{2} (z_1 - z_2)^\top \nabla^2 H(\tilde{z}) (z_1 - z_2), \quad (16)$$

where $\tilde{z} = z_2 + s(z_1 - z_2)$, $s \in [0, 1]$. Due to $\mathcal{G}_{A(t)}$ can be seen as an undirected and connected graph, the eigenvalues of its Laplacian

matrix $\mathcal{L}_{A(t)}$ are $0 = \lambda_1(\mathcal{L}_{A(t)}) < \lambda_2(\mathcal{L}_{A(t)}) \leq \dots \leq \lambda_n(\mathcal{L}_{A(t)})$. Taking the corresponding orthogonal eigenvectors $\mathbf{1}_{(n)}, e_2, \dots, e_n$ with $\|e_i\| = 1$, $i = 2, \dots, n$, then $y - y^*$ can be denoted as

$$y - y^* = \gamma_1 \mathbf{1}_{(n)} + \gamma_2 e_2 + \dots + \gamma_n e_n. \quad (17)$$

Let $r = \gamma_2 e_2 + \dots + \gamma_n e_n$, then $\|r\|^2 = \gamma_2^2 + \dots + \gamma_n^2$. Take $z_1 = y^*$ and $z_2 = y$ in (16), we have $H(y) - H(y^*) = \nabla H(y)^\top (y - y^*) - \frac{1}{2} (y^* - y)^\top \nabla^2 H(\tilde{y}) (y^* - y)$, where $\tilde{y} = y + s_1(y^* - y)$, $s_1 \in [0, 1]$. From the convexity of $H(y)$, it has $(y^* - y)^\top \nabla^2 H(\tilde{y}) (y^* - y) \geq 0$. Therefore, according to (14) and (17),

$$H(y) - H(y^*) \leq \nabla H(y)^\top (y - y^*) = \nabla F(x)^\top \mathcal{L}_{A(t)} r.$$

Because $\mathcal{L}_{A(t)}$ is positive semi-definite, there is $\mathcal{L}_{A(t)} = Q(t)^\top Q(t)$ for some $Q(t) \in \mathbb{R}^{n \times n}$, then

$$H(y) - H(y^*) \leq \nabla F(x)^\top Q(t)^\top Q(t) r \leq \|Q(t) \nabla F(x)\| \|Q(t) r\|. \quad (18)$$

Similarly, taking $z_1 = y$ and $z_2 = y^*$ in (16), then $\nabla H(y^*) = 0$ deduces that

$$H(y) - H(y^*) = \frac{1}{2} (y - y^*)^\top \nabla^2 H(\tilde{y}) (y - y^*),$$

where $\tilde{y} = y^* + s_2(y - y^*)$, $s_2 \in [0, 1]$. Based on (15), we have

$$H(y) - H(y^*) = \frac{1}{2} (y - y^*)^\top \mathcal{L}_{A(t)} \nabla^2 F(x) \mathcal{L}_{A(t)} (y - y^*).$$

From Assumption 1, there is $\kappa > 0$ satisfying $F(x)$ is κ -strongly convex. Hence, based on Lemma 2.1 and (17), we have

$$H(y) - H(y^*) = \frac{1}{2} (\mathcal{L}_{A(t)} r)^\top \nabla^2 F(x) (\mathcal{L}_{A(t)} r) \geq \frac{\kappa}{2} \lambda_2(\mathcal{L}_{A(t)}) \|Q(t) r\|^2. \quad (19)$$

Combining (18) with (19), it has

$$H(y) - H(y^*) \leq \frac{2}{\kappa \lambda_2(\mathcal{L}_{A(t)})} \|Q(t) \nabla F(x)\|^2. \quad (20)$$

For the distributed optimization algorithm (4), consider

$$V(y(t)) = \frac{1}{2} (H(y(t)) - H(y^*))^2. \quad (21)$$

As a result, the derivative of $V(y(t))$ with respect to time t is

$$\begin{aligned} \dot{V}(y(t)) &= (H(y(t)) - H(y^*)) \nabla H(y(t))^\top \dot{y}(t) \\ &= (H(y(t)) - H(y^*)) \eta(t)^\top \mathcal{L}_{A(t)} \dot{y}(t). \end{aligned} \quad (22)$$

Based on Lemma 2.1 and $\mathcal{L}_{A(t)} \mathbf{1}_{(n)} = 0_{(n)}$, it has

$$\begin{aligned} \eta(t)^\top \mathcal{L}_{A(t)} \dot{y}(t) &= -\eta(t)^\top \mathcal{L}_{A(t)}^2 \eta(t) + \eta(t)^\top \mathcal{L}_{A(t)} (g(t) \otimes \mathbf{1}_{(n)}) \\ &\leq -\frac{\lambda_2(\mathcal{L}_{A(t)})}{2} \eta(t)^\top \mathcal{L}_{A(t)} \eta(t). \end{aligned} \quad (23)$$

Thus, (20)–(23) and Lemma 3.2 imply that

$$\begin{aligned} \dot{V}(y(t)) &\leq -\frac{\lambda_2(\mathcal{L}_{A(t)})}{2} (H(y(t)) - H(y^*)) \eta(t)^\top \mathcal{L}_{A(t)} \eta(t) \\ &\leq -\frac{\kappa \lambda_2^2(\mathcal{L}_{A(t)})}{4} (H(y(t)) - H(y^*))^2 \\ &\leq -\frac{\kappa \lambda_2^2(\mathcal{L}_{A(0)})}{2} V(y(t)). \end{aligned} \quad (24)$$

To sum up, based on Lemma 2.2, one has $y(t)$ exponentially converges to the optimal solution to (13), which implies that $x(t)$ generated by the distributed optimization algorithm (4) converges to the optimal solution to VCDRA (2) exponentially. ■

Remark 4.2. The distributed optimization algorithm (4) proposed in this paper differs from the disturbance rejection distributed optimization algorithms presented in [25–27] in several aspects. Firstly, the distributed optimization algorithms provided in [25,26] are effective only for known external disturbances, which is unrealistic in most practical scenarios. On the contrary, the distributed optimization algorithm (4) can handle unknown external disturbances. Secondly, while the distributed optimization algorithm presented in [27] can handle unknown disturbances, it requires the external disturbance to

be constant to achieve an exponential convergence rate. In contrast, the distributed optimization algorithm (4) can achieve an exponential convergence rate under complicated external disturbances without selecting any control parameters. Finally, the distributed optimization algorithm (4) does not impose any boundedness assumption on the external disturbance $g(t)$, unlike [25–27], making it more applicable to a wider range of practical scenarios.

When the multi-agent system (1) is relaxed to be free of external disturbance, that is, $g(t) \equiv 0$, the distributed optimization algorithm (4) degenerates into

$$\begin{cases} \dot{x}_i(t) = \sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) + d_i \\ \dot{y}_i(t) = - \sum_{j=1}^n w_{ij}(t)(\eta_i(t) - \eta_j(t)) \end{cases} \quad (25)$$

where $\eta_i(t) = \frac{\partial}{\partial x_i} f_i(x_i(t), x_{\mathcal{N}_i^I}(t)) + \sum_{j \in \mathcal{N}_i^I(0)} \frac{\partial}{\partial x_j} f_j(x_j(t), x_{\mathcal{N}_j^I}(t))$, and $w_{ij}(t)$ is defined in (5). It is noteworthy that the distributed optimization algorithm (25) is fully distributed, relying solely on the private information of agent i in the system (1). In this case, the connectivity of the algorithm (25) under the interaction mechanism (5) can still be maintained, and its exponential convergence at the equilibrium point remains valid. Further details are provided in the following conclusion.

Corollary 4.1. Under Assumptions 1–2, if $\mathcal{G}_{W(0)}$ is connected and $K(0)$ is bounded, then $\mathcal{G}_{W(t)}$ is connected and $x(t) = \text{col}(x_i(t))$ generated by the distributed optimization algorithm (25) exponentially converges to the optimal solution $x^* = \text{col}(x_i^*)$ to VCDRA (2).

Proof. It is evident that the distributed optimization algorithm (25) represents a specific instance of algorithm (4) when $g(t) = 0$. Then recalling the definition of $K(t)$ in (7), it follows from (9), (10), and (12) that

$$\begin{aligned} & \dot{K}(t) \\ & \leq - \sum_{i,j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) - \sum_{r=1}^n w_{jr}(t)(\eta_j(t) - \eta_r(t)) \right) \\ & \quad + 2 \sum_{i=1}^n \left(\sum_{j=1}^n w_{ij}(t)(y_i(t) - y_j(t)) \right) \left(\sum_{r=1}^n w_{ir}(t)(\eta_i(t) - \eta_r(t)) \right) = 0. \end{aligned}$$

Similar to the analysis in Theorem 4.1, one has $\mathcal{G}_{W(t)}$ connected if $\mathcal{G}_{W(0)}$ is connected and $K(0)$ is bounded.

Following (21), the formulas (23) and (24) still hold, and thus $\dot{V}(y(t)) = -(H(y(t)) - H(y^*))\eta(t)^T \mathcal{L}_{A(t)} \eta(t) \leq -\kappa \lambda_2(\mathcal{L}_{A(0)}) V(y(t))$. As a result, the exponential convergence of the distributed optimization algorithm (25) can be proved easily by Theorem 4.2. ■

Remark 4.3. Based on the discussion of Theorems 4.1 and 4.2, it is evident that the strong convexity of the team cost function $F(x)$ in VCDRA (2) plays a substantial role in ensuring the convergence of the distributed optimization algorithm (4). Therefore, the algorithm proposed in this paper is suitable for the case that the team cost function is strongly convex but the private cost function may not be convex, which greatly broadens the application scope of the algorithm.

5. Numerical simulations

In this section, two applications in the fields of smart grid and wireless communication are presented to demonstrate the performance of the distributed optimization algorithm (4) for solving variable-coupled distributed resource allocations (VCDRA).

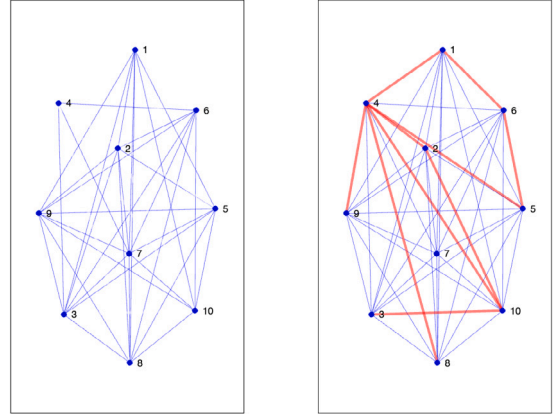


Fig. 3. Communication graph of the power network.

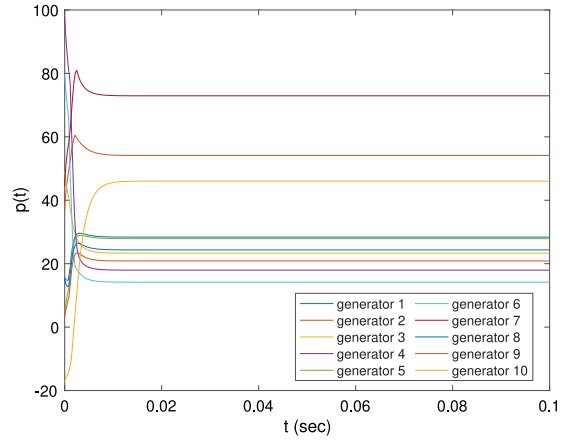
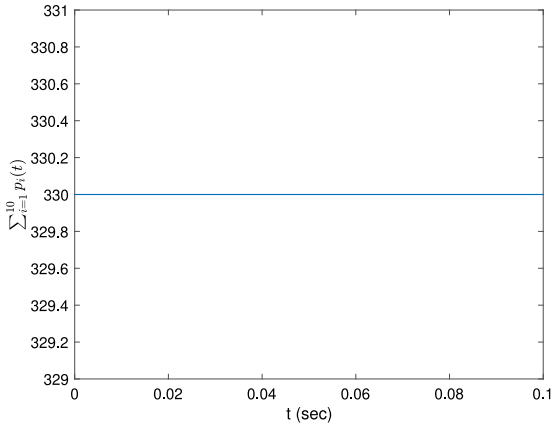
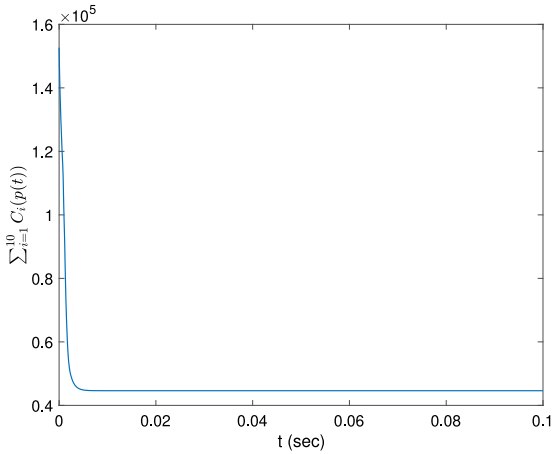
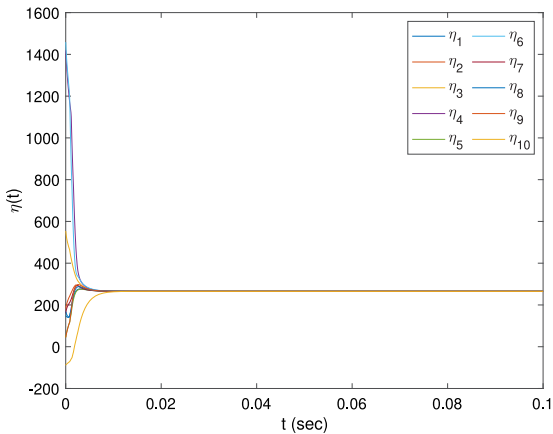


Fig. 4. Evaluation of the power references for the ten generators.

5.1. Power dispatch problem

The smart grid experimental design includes ten generators and a grid with a total power demand of 330, and the main goal of the power dispatch problem is to minimize the total transmission cost. In this experiment, we define the power demand as $\sum_{i=1}^{10} d_i = 330$, the power distribution variable of each generator as $p_i \geq 0$ ($i \in \mathcal{J}_{10}$), and its local transmission cost as C_i , that is, $C_1(p_1, p_5, p_{10}) = 4.6p_1^2 + 4.76p_1 + 0.05 \sin(p_5 - p_{10})$, $C_2(p_2, p_1, p_3) = 2.4p_2^2 + 5.56p_2 - 0.00021(p_1 + p_3)$, $C_3(p_3, p_6) = 5.6p_3^2 + 4.32p_3 + 0.0031\sqrt{1 + p_6^2}$, $C_4(p_4, p_3, p_7) = 7.2p_4^2 + 6.6p_4 + 0.0011\exp(-p_3/p_7)$, $C_5(p_5, p_1, p_7) = 4.7p_5^2 + 3.14p_5 - 0.0022(p_1 + p_7)^2$, $C_6(p_6) = 9.1p_6^2 + 7.54p_6$, $C_7(p_7) = 1.8p_7^2 + 3.28p_7$, $C_8(p_8, p_9) = 5.3p_8^2 + 7.31p_8 - 0.00451p_9$, $C_9(p_9, p_{10}) = 6.3p_9^2 + 2.45p_9 - 0.00351p_{10}$, and $C_{10}(p_{10}, p_1) = 2.8p_{10}^2 + 7.63p_{10} + \exp(-p_1)$. It is observed that $C_i(p)$ ($i \in \mathcal{J}_{10}$) may not necessarily be convex. However, the strong convexity of $C(p) = \sum_{i=1}^{10} C_i(p)$ for the power dispatch problem can be ensured by verifying its Hessian matrix. By taking the maximal interaction range $R = 5$, $\tau = 0.5$, and $y(0) = [2, 4, 5, 9, 1, 8, 5, 2, 1, -1]^T$, according to the interaction mechanism (5), the initial communication graph is shown as the left graph in Fig. 3 and it is obvious that Assumption 2 is satisfied.

To suppress the external disturbance $g(t) = 0.38 \sin t$, we apply the proposed distributed optimization algorithm (4) to minimize the team cost function $C(p) = \sum_{i=1}^{10} C_i(p)$ under the demand constraint $\sum_{i=1}^{10} p_i = \sum_{i=1}^{10} d_i$. As shown in Fig. 4, the evaluations of power supply by generators $i \in \mathcal{J}_{10}$ converge to the point $p^* = [28.38, 54.12, 23.31, 17.97, 27.89, 14.17, 72.93, 24.34, 20.87, 46.02]^T$. In Fig. 5, it implies that the total power supply always achieves the desired demand, and Fig. 6 illustrates the

Fig. 5. Evaluation of $\sum_{i=1}^{10} p_i(t)$.Fig. 6. Evaluation of $\sum_{i=1}^{10} C_i(p(t))$.Fig. 7. Evaluations of $\eta_i(t)$.

trajectory of the team cost value. According to Fig. 5 and the consensus of gradients η_i ($i \in \mathcal{J}_{10}$) shown in Fig. 7, it follows that the obtained p^* is the optimal power supply of the power dispatch problem in this example. Thus, the effectiveness of the distributed optimization algorithm (4) in handling VCDRA with the variable-coupled private cost functions can be guaranteed. Additionally, the distances of the states $y_i(t)$ of agents are illustrated in Fig. 8. As observed in Fig. 8, for any link $(i, j) \in \mathcal{E}_{W(0)}$, it holds $|y_i(t) - y_j(t)| < R$ and several new links

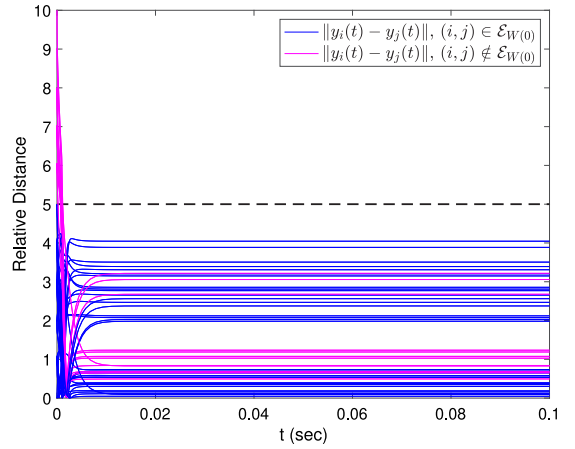


Fig. 8. Communication links.

Table 1

The comparison of convergence time when $g(t) = 0$.

	Algorithm (4) from $y(0)$	Algorithm (4) from $y'(0)$	Algorithm in [20] from $y'(0)$
Convergence time	0.005 s	0.01 s	0.04 s

have emerged, which supports the connectivity-preserving property of the interaction mechanism (5).

Note that the disturbance $g(t)$ considered in this paper can be unbounded, we retake it in this example as $g(t) = 0.38t \sin t$, which is shown in Fig. 9(a). Through applying the distributed optimization algorithm (4) to solve the above VCDRA with the same parameters, the evolution of $p_i(t)$, the gradient $\eta_i(t)$ and the communication links of the power network can be seen in Figs. 9(b), 9(c), and 9(d), respectively. To provide a more comprehensive demonstration of the algorithm's effectiveness, the state trajectory generated by the algorithm (4) from the initial state $y(0)$ without external disturbances, i.e. $g(t) = 0$, is shown in Fig. 10 (Solid line). Combined with Figs. 4, 9(b), and 10, it is concluded that the distributed optimization algorithm (4) can also effectively suppress more general disturbance and obtain the optimal solution of the VCDRA.

In addition, by resetting the initial state to zero (that is $y'(0) = 0_{(10)}$), the algorithm (4) proposed in this paper and the algorithm for processing VCDRA in [20] are respectively applied. The results can be seen from Fig. 10. Table 1 can be obtained from Figure 10. It intuitively shows the convergence time in each case, and the results show that the convergence time of algorithm (4) is faster, which is 75% shorter than that of the algorithm proposed in [20]. Also, Fig. 10 implies that the validity of the distributed optimization algorithm (4) is independent of the initial value. Fig. 11 displays the state trajectory generated by the algorithm in [20] under unbounded external disturbance. Obviously, the proposed algorithm (4) demonstrates superior robustness and convergence performance.

5.2. Spectrum resource allocation

In order to complete global communication, global navigation, global environmental monitoring, and other tasks, various countries are stepping up the study of the global satellite communication network. Using constellations to provide a variety of broadband multimedia services for the world is also the development trend of future communication satellite systems. Constellation refers to a network of satellites composed of several satellite rings arranged in a certain way. In recent years, the navigation satellite system has also proposed to use the inter-satellite link (ISL) and on-board processing technology to realize the

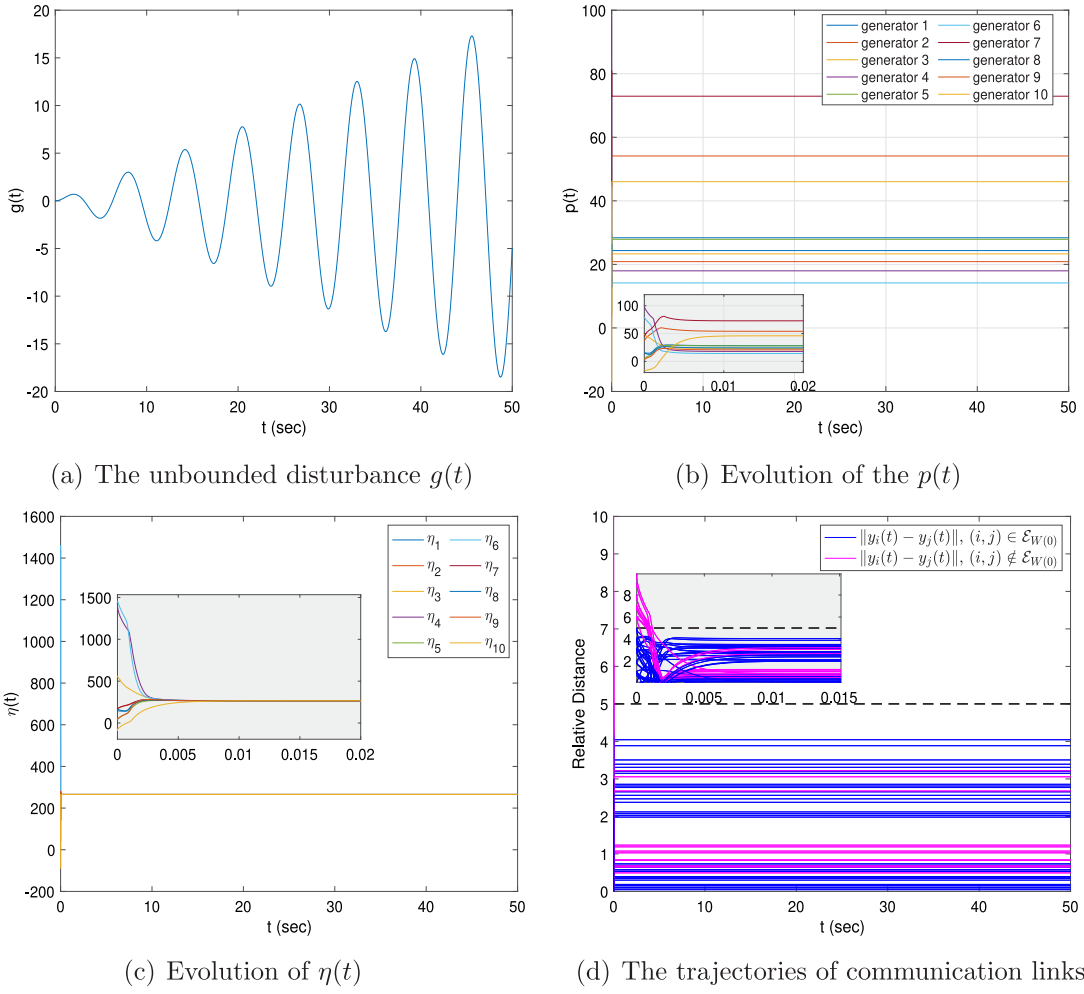


Fig. 9. Evaluations of $g(t)$, $p(t)$, $\eta(t)$, and $|y_i(t) - y_j(t)|$ under the unbounded disturbance.

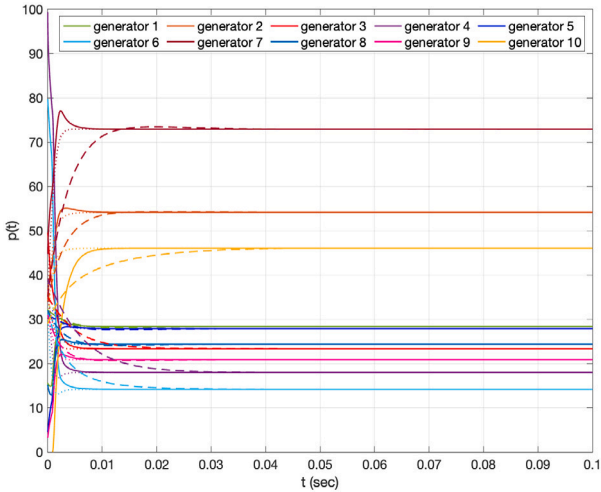


Fig. 10. Trajectories of $p(t)$ with $g(t) = 0$. Dashed lines: the optimization algorithm (4) from $y(0)$. Solid lines: the optimization algorithm in [20] from $y'(0)$. Dotted lines: the optimization algorithm (4) from $y'(0)$.

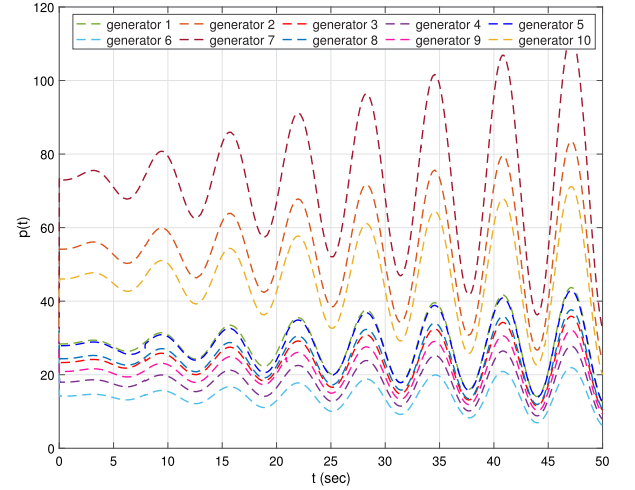


Fig. 11. Evolution of $p(t)$ of the optimization algorithm in [20] with $g(t) = 0.38t \sin t$.

mutual ranging and data exchange between satellites, aiming to achieve satellite autonomous navigation [24].

Taking the Galileo satellite constellation as an example, it is composed of 30 satellites evenly distributed in three orbits, with ten satellites on each orbital plane and one serving as a backup. Here, we assume that the three orbital planes are located in low-earth orbit (LEO, $100 \text{ km} < h < 2000 \text{ km}$), medium-earth orbit (MEO, $2000 \text{ km} < h <$

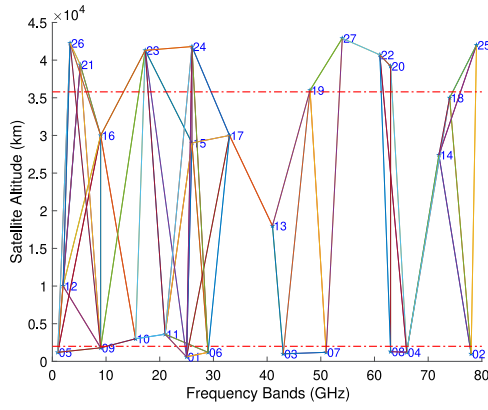


Fig. 12. Layout and initial connection of the Galileo satellite.

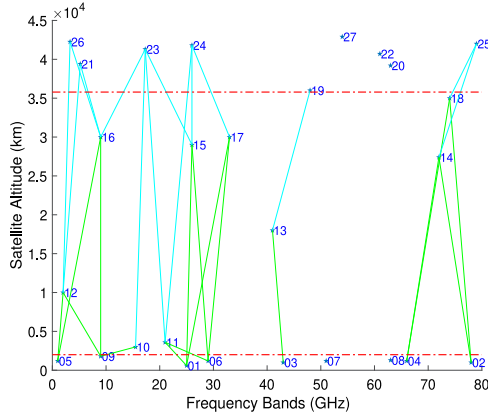


Fig. 13. Interaction relation of the Galileo satellite constellation.

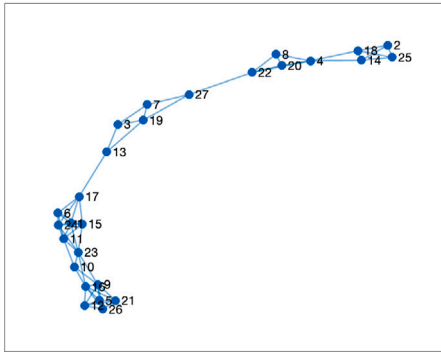


Fig. 14. Initial communication network of the Galileo satellite constellation.

35786 km) and high-earth orbit (HEO, $h > 35786$ km), respectively, and all 27 satellites can work normally.

In general, satellite links can be divided into intra-layer satellite links, inter-layer satellite links, and satellite-ground links. Intra-layer satellite links include links within the same orbital plane and links within different orbital planes. Inter-layer satellite links include links between LEO layer satellites, MEO layer satellites, and HEO layer satellites. These two are collectively referred to as ISL. To improve the autonomy of the satellite network, the satellite system adopts the ISL technology, so that there are direct communication channels between satellites.

In this case, the distributed spectrum resource allocation of the Galileo satellite constellation is taken into account. The objective is to reduce the signal transmission loss as much as possible under the

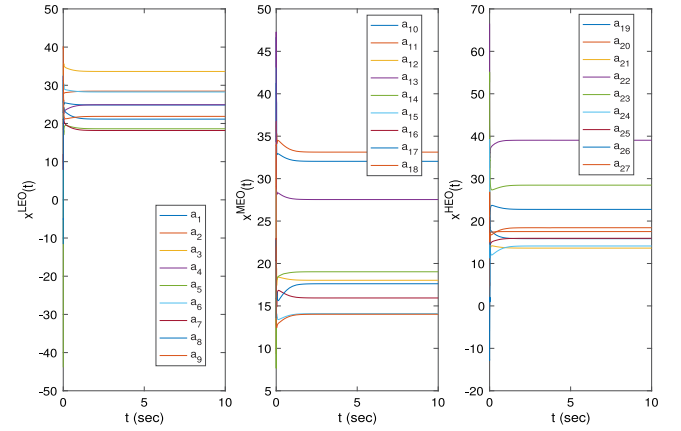


Fig. 15. Evaluation of the electromagnetic signal intensity of 27 satellites.

premise of electromagnetic signal feedback between satellites in adjacent layers and normal operation of various radio services, which can be represented as the following VCDRA

$$\min \sum_{i=1}^{27} f_i(x) \quad \text{s.t.} \quad \sum_{i=1}^{27} x_i = d, \quad (26)$$

where $f_i(x)$ and x_i are the signal transmission loss and the electromagnetic signal intensity of satellite i , respectively, and d is the total user demand.

Assume that the initial frequency bands of satellites are $y(0) = [25, 78, 43, 66, 1, 29, 51, 63, 9, 15.5, 21, 2, 41, 72, 26, 9, 33, 74, 48, 63, 5.2, 61, 17.3,$

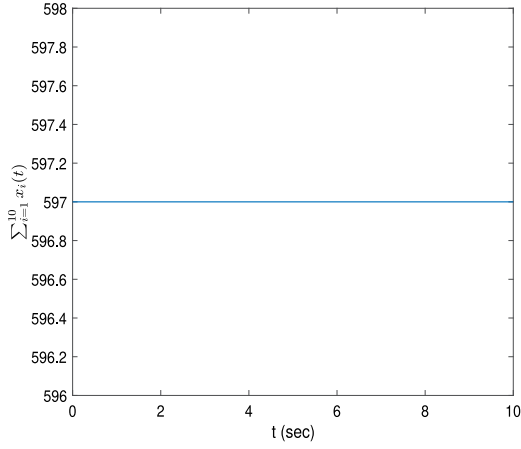
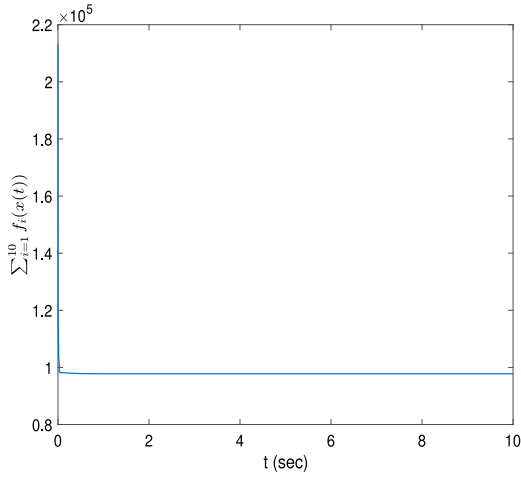
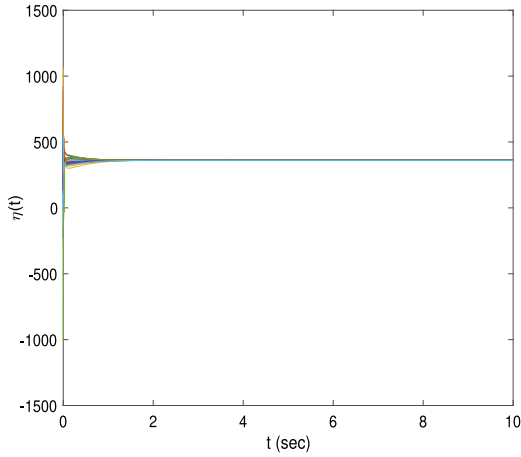
$26, 79, 3.3, 54]^T$ (GHz), since the connectivity of the link is determined by coverage range of electromagnetic signals between satellites, namely satellites a_i and a_j can communicate only when $|y_i - y_j| < R$ ($R = 9$ GHz is the accessible range). Hence, the communication network of the Galileo satellite constellation is shown in Fig. 12. According to the altitude of satellites and the mutual influence of electromagnetic signals between satellites, the electromagnetic signal influence of satellites between adjacent layers is considered here, as shown in Fig. 13. It is found from Figs. 12 and 13 that $R = 9$ is appropriate such that Assumption 2 is satisfied. In addition, Fig. 14 is the abstract topology of the communication network in Fig. 12, which shows that the initial communication network is an undirected and connected topology.

Let $x = \text{col}(x_i)_{i=1}^{27}$ be the electromagnetic signal intensity of the Galileo satellite constellation, and $d = 597$ be the total user demand. According to [44,45], the signal transmission loss of the satellite a_i is expressed as

$$f_i(x) \triangleq f_i(x_i, x_{\mathcal{N}_i^I}) = \alpha_i x_i^2 + \beta_i x_i + \sum_{j \in \mathcal{N}_i^I} k_{i,j} x_i x_j,$$

where $\alpha_i > 0$, $k_{i,j}$ is the influence coefficient of satellite a_j on the signal transmission loss of satellite a_i , and $\mathcal{N}_i^I$ is the set of all satellites that interact with satellite a_i . Taking satellite a_{17} in MEO as an example, it can be seen from Fig. 13 that $\mathcal{N}_{17}^I = \{1, 6, 24\}$, thus the signal transmission loss function of satellite a_{17} is $f_{17}(x) = \alpha_{17} x_{17}^2 + \beta_{17} x_{17} + \sum_{j \in \mathcal{N}_{17}^I} k_{17,j} x_{17} x_j$, which obviously depends on the electromagnetic signal strengths of satellites a_1 , a_6 in LEO, and a_{24} in HEO.

To illustrate the effectiveness of the algorithm (4), the coefficients $\alpha_i \in [1, 15]$ and $\beta_i \in [-50, 0]$ ($i \in \mathcal{J}_{27}$) are determined randomly in the VCDRA (26). Let $k_{i,j} = 0.02(i + j)$, and the external unknown interference in the multi-agent system (1) be $g(t) = \cos 2t$, it is clear that $F(x) = \sum_{i=1}^{27} f_i(x)$ is strongly convex. Therefore, the trajectories of electromagnetic signals $x^{\text{LEO}} = \text{col}(x_i)_{i=1}^9$, $x^{\text{MEO}} = \text{col}(x_i)_{i=10}^{18}$, and $x^{\text{HEO}} = \text{col}(x_i)_{i=19}^{27}$ over time can be obtained by applying the distributed optimization algorithm (4), and are shown in Fig. 15. From Figs. 16 and 17, it can be verified that the total supply of

Fig. 16. Evaluation of $\sum_{i=1}^{27} x_i(t)$.Fig. 17. Evaluation of $\sum_{i=1}^{27} f_i(x(t))$.Fig. 18. Evaluation of $\eta(t)$.

electromagnetic signals of the Galileo satellite constellation can meet the user demand and achieve the minimum consumption. Combining Figs. 16 and 18, it can deduce that $x(t)$ converges to the optimal solution $x^* = [24.84, 28.46, 33.62, 24.91, 18.61, 28.25, 18.16, 21.14, 21.86, 32.04, 33.12, 18.02, 27.54, 19.02, 14.09, 15.94, 17.61, 14.02, 15.85, 17.52, 13.63, 39.05, 28.46, 14.12, 15.94, 22.75, 18.43]^T$ of the VCDRA (26), which

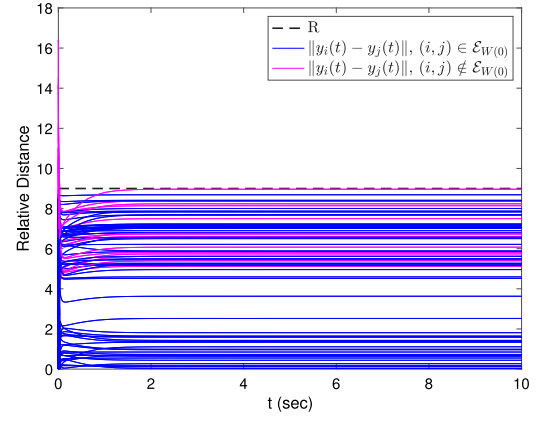


Fig. 19. The trajectories of communication links.

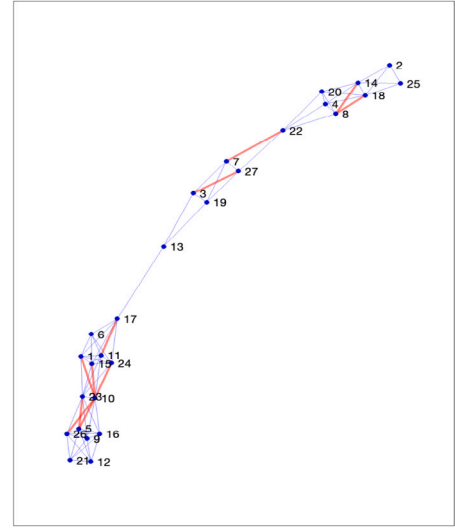


Fig. 20. The steady communication network of the Galileo satellite constellation.

reflects the effectiveness of the distributed optimization algorithm (4). In addition, Fig. 19 demonstrates the connectivity-preserving property of the proposed interaction mechanism (5), and the communication network of the Galileo satellite constellation is displayed in Fig. 20. To sum up, the numerical results provide evidence for the theoretical results obtained in this paper.

6. Conclusion

Building upon multi-agent systems constrained by admissible interaction ranges and influenced by external disturbances, this paper has introduced a new communication mechanism and developed a novel distributed optimization algorithm. The algorithm has been mathematically proven to converge exponentially to the optimal solution for distributed resource allocation problems with variable-coupled private cost functions. Considering the controller design and system dynamics, the proposed distributed optimization algorithm has achieved complete distribution and ensured effectiveness without relying on known disturbance bounds, distinguishing it from most existing distributed optimization algorithms.

Simulations in smart grid and wireless communication applications have validated the algorithm's performance. The results have demonstrated the designed distributed optimization algorithm's robustness against unbounded external disturbances and its effectiveness for VCDRAs. In addition, the algorithm in this paper has a competitive

convergence speed. These metrics underscore the algorithm's practical applicability and superiority in handling complex, real-world scenarios.

It should be noted that the external disturbances considered in this paper are consistent for all agents, and local constraints or coupled inequality constraints have not been involved. Future research could extend the proposed distributed optimization algorithm to achieve enhanced performance and handle more complex constraints.

CRedit authorship contribution statement

Linhua Luan: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Sitian Qin:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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